

AI Hallucination Courtroom Classroom Run Kit

TOK Cognitive Science Activity Bank • Knowledge and Technology • 35-45 min

Category	Knowledge and Technology	Time	35-45 min
Difficulty	Extended	ID	ai-hallucination-courtroom

Big idea: Fluency can be mistaken for knowledge.

One-Minute Teacher Brief

Students practise evaluating fluent but unreliable AI-generated claims.

Student Prompt

Put the AI paragraph on trial. For each claim, decide: grounded, unsupported, false, or needs verification.

Concrete Stimulus Packet

AI Paragraph On Trial	Students should verify AI-generated answers because large language models do not truly understand facts; they predict likely sequences of words based on patterns in training data. This means they can produce answers that sound fluent and authoritative even when details are wrong. For example, a 2021 study by Elena Marsh and David Kline in the Journal of Digital Reasoning found that 68% of AI-generated school summaries contained at least one factual error. AI tools are also connected to the live internet by default, so they usually update themselves instantly when new information appears. Because of this, students should treat AI as a starting point for inquiry, not as a final authority.	Prompt given to AI: Explain why students should verify AI-generated answers.
Courtroom Roles	Prosecution: argue the paragraph is unreliable. Defence: argue parts are useful if treated carefully. Expert witness: explain hallucinations. Judge: classify claims. Jury: vote mostly reliable, partly reliable, or misleading.	Assign roles before students see the hidden issues.
Hidden Issues	Fabricated citation: Elena Marsh and David Kline, Journal of Digital Reasoning, 2021. Plausible false detail: AI tools are connected to the live internet by default. Grounded claim: AI can produce fluent but incorrect answers. TOK point: fluency can create misplaced trust.	Teacher reveal after verdicts.
Claim Check: AI predicts likely word sequences	Verdict: Grounded	This broadly describes how language models generate text.
Claim Check: AI can sound authoritative while wrong	Verdict: Grounded	This is a known risk of generative AI.
Claim Check: Marsh and Kline study, 2021	Verdict: False/Fabricated	The citation is invented for this activity.

Claim Check: 68% of school summaries had errors	Verdict: Unsupported	Depends entirely on the fabricated study.
Claim Check: AI tools are live online by default	Verdict: False	Some tools can browse, but many do not by default.
Claim Check: AI should be a starting point, not final authority	Verdict: Reasonable conclusion	Supported by the earlier risks.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	Which claim in the paragraph sounds most authoritative?	The fabricated 2021 study and percentage often sound authoritative.
2	Which claim can be accepted as broadly grounded?	AI can produce fluent but incorrect answers; language models predict likely word sequences.
3	Which claim is fabricated?	The Marsh and Kline citation in the Journal of Digital Reasoning.
4	Which claim is plausible but false or overgeneralised?	AI tools are connected to the live internet by default.
5	What made the paragraph feel trustworthy?	Fluent tone, specific names, journal title, percentage, and confident explanation.
6	Can this paragraph count as knowledge? Why or why not?	Partly; some claims are reasonable, but the fabricated/unsupported claims prevent treating it as reliable knowledge without verification.

Student Task

For each claim, decide: grounded, unsupported, false, or needs verification. Then write: “Can this paragraph count as knowledge? Why or why not?”

Teacher Key / Reveal

The paragraph is partly useful but not reliable as knowledge without verification because it combines grounded claims with fabricated and overgeneralised details.

- AI predicts likely word sequences: Grounded — This broadly describes how language models generate text.
- AI can sound authoritative while wrong: Grounded — This is a known risk of generative AI.
- Marsh and Kline study, 2021: False/Fabricated — The citation is invented for this activity.
- 68% of school summaries had errors: Unsupported — Depends entirely on the fabricated study.
- AI tools are live online by default: False — Some tools can browse, but many do not by default.
- AI should be a starting point, not final authority: Reasonable conclusion — Supported by the earlier risks.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.

- Use the debrief to answer: Can fluency be mistaken for knowledge?

Setup Checklist

- Prepare a starter stimulus using: An AI-generated paragraph with one fabricated citation and one plausible false detail.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- An AI-generated paragraph with one fabricated citation and one plausible false detail.
- Courtroom roles for prosecution, defence, expert witness, judge, and jury.
- An evidence table asking which claims are grounded, unsupported, or unverifiable.

Resources To Use

- Resource pack: <resources/ai-hallucination-courtroom/index.html>
- Card deck CSV: <resources/ai-hallucination-courtroom/cards.csv>
- Visual prompt: <resources/ai-hallucination-courtroom/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Give all groups the same paragraph.
2. Assign roles: prosecution, defence and judge.
3. Prosecution identifies unsupported or false claims. Defence argues how the tool could still be useful. Judges decide what evidence is required.
4. Finish by writing class rules for responsible AI use.

Debrief Moves

- Knowledge question: Can fluency be mistaken for knowledge?
- Can a machine be a knower, or only a producer of outputs?
- Who is responsible for AI misinformation?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...